

Science Has an Objective Function

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Abstract

Science has always had an objective function. It has never been written down in a single place. Written down, it is: minimize $L(H) + L(D | H)$, where H is a hypothesis, D is observed data, $L(H)$ is the description length of the hypothesis (approximating its Kolmogorov complexity), and $L(D | H)$ is the description length of the data given the hypothesis (the negative log-likelihood). This is the Minimum Description Length (MDL) principle—the computable approximation to Solomonoff induction, which is Bayesian inference with the Kolmogorov/Solomonoff prior. We argue that this framework is not a new proposal but a formal statement of what science has always been doing informally. We develop the framework from first principles, trace the history of philosophy of science as a sequence of partial approximations to it, and identify ten cases where ideas from different traditions—philosophy, computability theory, information theory, neuroscience, machine learning—turn out to be the same idea, expressed in different language. We close with the observation that two current developments—autonomous AI research systems and machine-verified formal proof—are direct engineering instantiations of the framework, and that the tools now exist to approximate the ideal at scale.

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1 The Objective Function

1.1 The Claim

Science finds short descriptions of large bodies of observations. The right way to formalize “short description” is Kolmogorov complexity. The right way to formalize “finds” is Bayesian inference with the Solomonoff prior. Everything else in this paper is unpacking those three sentences.

The formal statement. Let H be a hypothesis—a theory, a model, a program—and let D be observed data. The objective function for science is:

$$\text{maximize} \quad \log P(H \mid D) = \log P(D \mid H) + \log P(H) + \text{const} \quad (1)$$

by Bayes’ rule, where $P(D \mid H)$ is the likelihood and $P(H)$ is the prior. This is uncontroversial as a framework. The only question is: what prior?

1.2 The Prior Problem

If the prior $P(H)$ can be chosen arbitrarily, Bayes’ rule is a framework, not a complete theory. Two scientists with different priors will reach different posteriors from the same data. The standard objection to Bayesianism is that this makes science subjective.

The Kolmogorov/Solomonoff answer: the prior is not arbitrary. There is a unique, principled choice grounded in the theory of computation.

1.3 The Solomonoff Prior

The Kolmogorov complexity $K(x)$ of a string x is the length of the shortest program (on a fixed universal Turing machine U) that outputs x and halts:

$$K(x) = \min\{|p| : U(p) = x\}.$$

Solomonoff [14] proposed assigning to each hypothesis H the prior:

$$P(H) \propto 2^{-K(H)}.$$

Simpler hypotheses—those with shorter generators—receive exponentially higher prior probability. This is Occam’s Razor, derived from first principles rather than assumed as a philosophical preference.

The Solomonoff prior is *universal*: for any computable probability distribution P over strings, there exists a constant c_P such that $M(x) \geq c_P \cdot P(x)$ for all strings x , where M is the Solomonoff measure. No computable prior assigns more probability to any hypothesis than M does, up to a fixed factor. The prior dominates every computable alternative.

1.4 Minimum Description Length

Because K is uncomputable (Section 2), the practical version is the Minimum Description Length (MDL) principle [13]. Instead of the shortest program, use the shortest description under a fixed code. The objective becomes:

$$\text{minimize} \quad L(H) + L(D \mid H) \quad (2)$$

where $L(H)$ is the description length of the hypothesis and $L(D | H)$ is the description length of the data given the hypothesis (the negative log-likelihood under a specific coding scheme).

The two terms trade off directly. $L(H)$ penalizes complexity: a theory that is too elaborate spends too many bits describing itself. $L(D | H)$ penalizes poor fit: a theory that predicts the data badly leaves many residual bits to transmit. The optimum is the theory that compresses the combined description most. Science is the search for short two-part codes for the world.

1.5 Occam’s Razor, Derived

A consequence worth stating explicitly: two theories that predict the data equally well are not equally good. The simpler one has higher posterior. A more complex theory must predict the data *better* to compensate for its higher complexity penalty. The scientist who adds epicycles to save a theory is doing something formally wrong: increasing $K(H)$ without a commensurate gain in $\log P(D | H)$.

The objective function is *objective* in both senses of the word: it is a target (something to minimize) and it is observer-independent (grounded in the structure of computation, not in any particular scientist’s preferences). The Solomonoff prior is invariant across observers up to a fixed constant determined by the choice of universal Turing machine—not by whoever is doing the science.

2 Kolmogorov Complexity

2.1 The Intuition

Consider two strings of a thousand bits. String A alternates 0 and 1 indefinitely; String B is a typical output of a fair coin. Both are a thousand bits long. String A can be described in a few dozen characters: “alternating 0 and 1, one thousand times.” String B, if genuinely random, cannot be described more briefly than writing it out in full.

Kolmogorov complexity makes this precise. The complexity $K(x)$ of a string x is the length of the shortest computer program that outputs x and halts. String A has very low K ; String B has K close to its own length.

Three examples locate the concept concretely. The digits of π look random but have very low K : any standard algorithm for computing π provides a program of a few hundred bytes that generates arbitrarily many digits. Newton’s law of universal gravitation $F = Gm_1m_2/r^2$ is a program of a few dozen bytes that predicts the trajectories of every planet, moon, and comet we have observed, to many decimal places. A random genome, by contrast, would have K close to its raw length—but the actual human genome is substantially more compressible, because evolution is a compressive process.

2.2 The Formal Definition

Fix a universal Turing machine U . The Kolmogorov complexity of a finite binary string x with respect to U is:

$$K_U(x) = \min\{|p| : U(p) = x\}$$

where $|p|$ is the length of program p in bits. In practice, the prefix-free variant (where no valid program is a prefix of another) is standard; it has cleaner information-theoretic properties.

The key theorems:

Theorem 2.1 (Invariance [7]). *For any two universal Turing machines U and V :*

$$|K_U(x) - K_V(x)| \leq c_{U,V}$$

where $c_{U,V}$ depends only on the machines, not on x . For large strings, this constant is negligible relative to $K(x)$.

The Invariance Theorem is what makes $K(x)$ objective: the choice of machine changes the value by at most a fixed overhead, independent of the string being measured. For any two computable descriptions, the complexity is the same up to a translator constant.

Proposition 2.2. *For any length n , at most 2^{n-c} strings of length n have $K(x) < n - c$. Most strings are incompressible.*

This is a counting argument: there are only 2^{n-c} programs shorter than $n - c$ bits. When we encounter a compressible string in nature—a genome, a physical trajectory, a spectrum—that compressibility is evidence of an underlying generative process. Structure is rare.

2.3 Uncomputability: A Feature, Not a Bug

$K(x)$ is not computable. No program, given x as input, reliably outputs $K(x)$ and halts. The proof is a formalization of Berry’s paradox: supposing K were computable, one could construct a short program that outputs a string with high complexity—a contradiction.

This is sometimes presented as a defect. It is not. Kolmogorov complexity is the *right* measure of structural complexity. Its uncomputability is the signature of universality: K captures structure defined by all possible programs, not any particular one. If K were computable, it would be reducible to some specific algorithm, and that algorithm would make implicit choices about what structure counts.

The scientist’s relationship to K is the same as the mathematician’s relationship to real numbers: most are uncomputable, yet the reals are the right framework. Scientists approximate K from above. Every proposed theory is an upper bound on $K(D)$. The search for shorter descriptions is the search for a tighter bound. The fact that the true minimum is uncomputable is precisely why the search never ends.

2.4 MDL as the Computable Approximation

The Minimum Description Length principle [13] replaces K with a description length under a fixed coding scheme. For a parametric model H with parameters θ :

$$L(H) + L(D | H) = L(\theta) - \log P(D | \theta)$$

This is computable for any fixed model class and coding scheme. MDL is model-selection consistent: as data grows, it converges to the true model when the true model is in the hypothesis class. It reduces overfitting by penalizing complexity directly, without ad hoc regularization.

The relationship to Bayes is exact. Under a prior $P(H) \propto 2^{-L(H)}$:

$$\log P(H | D) = -L(H) + \log P(D | H) + \text{const} = -\text{MDL}(H, D) + \text{const}$$

Maximizing the Bayesian posterior is equivalent to minimizing MDL, for the right choice of prior and coding scheme.

2.5 Connections to Information Theory

The expected Kolmogorov complexity of a draw from a computable distribution P equals its Shannon entropy $H(P)$ up to lower-order terms:

$$\mathbb{E}[K(X)] \approx H(X).$$

Shannon entropy is Kolmogorov complexity averaged over a distribution. Kolmogorov complexity is the pointwise version: it applies to individual strings, without requiring a distribution. When you have a single string and want to know how much structure it has—without committing to a probability model—Kolmogorov complexity is the right tool.

Lossless compression (gzip, bzip2, zstd) provides empirical upper bounds on K . A string that compresses well has low K ; one that does not compress has high K . This means compression ratio is a practical proxy for structural complexity, usable without specifying any explicit model.

3 The History of Philosophy of Science as a Single Argument

The history of philosophy of science is a single long argument about one problem: Hume’s problem of induction. Every major movement is either a response to it, a refinement of a previous response, or an attempt to dissolve it. The Bayesian/Kolmogorov framework is where the argument arrives. We trace the through-line here, briefly; each movement clarifies what the framework is solving.

3.1 Hume: The Impossibility

David Hume [5] identified the deepest problem in the philosophy of science. The problem of induction: no finite set of observations logically justifies a universal law. The inference from “all observed F s are G s” to “all F s are G s” is not deductively valid. You can observe a million white swans; the next one might be black. It was, in Australia.

Worse: the obvious response—induction is justified because it has worked in the past—is itself an inductive argument. The justification is circular. Hume concluded that inductive inference is a matter of psychological habit, not reason.

3.2 Popper: The Right Asymmetry, the Wrong Conclusion

Karl Popper [10] noticed that confirmation is logically weak while falsification is logically strong. No number of white swans proves the law; one black swan refutes it. He proposed falsifiability as the demarcation criterion for science and characterized scientific progress as conjecture and refutation: bold hypotheses, then honest attempts to falsify them.

Popper was right about the asymmetry and right that unfalsifiable theories are scientifically empty. He was wrong to conclude that science involves no induction. His “corroboration” measure—how well a theory has survived tests—is, precisely defined, the Bayesian likelihood ratio $P(E | H)/P(E | \neg H)$: the update factor, stripped of the prior he refused to use. Popper computed the right quantity and declined to multiply by the prior. The Solomonoff prior completes his programme.

3.3 Kuhn: The Right Phenomenology, No Normative Theory

Thomas Kuhn [8] argued that Popper’s account does not match how science actually works. Scientists operate within *paradigms*—shared frameworks of assumptions and methods—and do not abandon them at the first anomaly. When anomalies accumulate to the point where confidence breaks down, a revolution occurs: one paradigm replaces another, discontinuously.

Kuhn correctly described the phenomenology. He lacked a normative account: when *should* a paradigm shift occur? The MDL framework supplies the answer (Section 4).

3.4 Lakatos: The Right Criterion, Incompletely Formalized

Imre Lakatos [9] synthesized Popper and Kuhn via the methodology of scientific research programmes. Each programme has a *hard core* of protected assumptions and a *protective belt* of auxiliary hypotheses that absorb anomalies. A programme is *progressive* if its protective belt modifications generate new confirmed predictions; *degenerating* if they only explain known anomalies without predicting new ones.

Under MDL, a progressive programme is one where successive modifications decrease $L(H) + L(D \mid H)$ on new data. A degenerating programme is one where modifications increase $L(H)$ without decreasing $L(D \mid H)$ on new observations. Lakatos gave the right criterion; MDL makes it computable.

3.5 Feyerabend: Methodological Anarchism as the Solomonoff Prior

Paul Feyerabend [2] argued that no hypothesis should be ruled out *a priori*: science has always advanced by entertaining ideas that violated current methodological rules. “Anything goes” as a prior policy.

The Solomonoff prior assigns nonzero probability to every computable hypothesis. Nothing is excluded before the data speaks. Feyerabend’s permissiveness is not anarchism—it is the correct prior policy. A prior that excludes any computable hypothesis is provably suboptimal. What Feyerabend called anarchism, Solomonoff called universality.

3.6 Bayesian Confirmation: The Culmination

Bayesian confirmation theory [4] provides the most complete response to Hume. Evidence E confirms H if and only if $P(H \mid E) > P(H)$; the degree of confirmation is the likelihood ratio $P(E \mid H)/P(E)$. This dissolves the raven paradox (observing a red fire engine confirms “all ravens are black,” but only infinitesimally), handles old evidence via counterfactual reasoning, and validates Lakatos’s distinction as a distinction between increasing and decreasing marginal likelihood.

Carnap spent his career trying to construct an inductive logic—a formal system that would determine the degree of confirmation of any hypothesis by any evidence. He never succeeded because he had no principled account of the prior. The Solomonoff prior completes the programme he started.

4 Ten Identifications

The directories of the `science` repo were built separately. This section is about what happens when those threads cross. The cases below are not analogies—loose, suggestive comparisons. They are *identifications*: cases where two ideas from different traditions turn out to be the same idea, expressed in different language, developed by people who were not talking to each other. The convergence is evidence that something real is being pointed at.

4.1 Hume’s Problem Is the Halting Problem

Hume showed that induction cannot be logically justified: you cannot know, from finite observations, that a universal law holds. Turing showed that you cannot decide, in general, whether a program halts.

These are the same result. A universal law is a short program—a generator for an infinite class of observations. Confirming the law means certifying that the proposed program is the shortest consistent with the data. But certifying that no shorter program exists requires checking whether every shorter program eventually fails to match the data—instances of the halting problem. Since the halting problem is undecidable, $K(x)$ is uncomputable, and induction cannot be deductively justified.

The reason you cannot justify induction is the same reason $K(x)$ is uncomputable. Hume’s problem is not a philosophical puzzle awaiting a philosophical solution. It is an undecidability result, and it has the same status as the halting problem: provably unsolvable, not merely unsolved.

What can be done—and what the Bayesian/Kolmogorov framework does—is show that the impossibility of certainty does not prevent optimal inference. Solomonoff induction is provably optimal for prediction in any computable environment. The gap between “optimal” and “certain” is exactly the gap between computable approximations to $K(x)$ and $K(x)$ itself.

4.2 Paradigm Shifts Are MDL Phase Transitions

Kuhn described paradigm shifts as discontinuous and partly irrational. Under MDL, they are mathematically inevitable and perfectly rational.

A paradigm is a hypothesis class—a family of theories sharing common structure. Normal science is optimization within the class: adjusting parameters, adding auxiliary hypotheses, refining the model. Each adjustment is legitimate as long as it decreases total description length.

A paradigm shift occurs when a competing hypothesis class achieves lower $L(H) + L(D | H)$ than the incumbent with all its protective belt modifications. The shift is discontinuous because hypothesis classes are discrete objects: you cannot smoothly interpolate between Ptolemaic and Copernican astronomy. But it is not irrational. The new paradigm wins because it compresses the data better.

The discontinuity is a phase transition in the physics sense: a sharp change in the optimal state of the system (which paradigm minimizes description length) as a continuous parameter (accumulated data) crosses a threshold. Kuhn correctly identified the phenomenology of phase transitions in scientific knowledge. MDL provides the thermodynamics underneath.

4.3 Popper’s Corroboration Is Bayesian Updating Without the Prior

Popper’s corroboration measure, precisely defined [11], is:

$$C(H, E) = \frac{P(E | H) - P(E)}{P(E | H) - P(E \cdot H) + P(E)}$$

This is a normalized likelihood ratio: how much more probable E is under H than under the background probability of E . It is purely a function of likelihoods. It does not involve the prior $P(H)$.

In Bayesian terms, Popper was computing exactly half of the update:

$$P(H | E) \propto \underbrace{P(E | H)}_{\text{Popper's corroboration}} \cdot \underbrace{P(H)}_{\text{refused}}$$

He refused the prior because he found subjective probability unacceptable. He was right to find it unacceptable; he was wrong to conclude the prior must be abandoned. The Solomonoff prior $P(H) \propto 2^{-K(H)}$ is objective: a property of the hypothesis itself, invariant across observers. With it, corroboration becomes posterior probability, and the refusal of induction dissolves.

4.4 The Replication Crisis Is an MDL Accounting Error

The replication crisis in psychology and social science resulted from p-hacking: testing many hypotheses and reporting only those achieving $p < 0.05$. This is an MDL accounting error.

When a researcher tests N hypotheses and reports the best, the true description length of the reported hypothesis is not $L(H_{\text{reported}})$. It is:

$$L(H_{\text{reported}}) + \log_2 N$$

The extra $\log_2 N$ bits is the cost of the search: the information required to specify which of the N tested hypotheses was selected. Omitting this cost produces apparent results that do not replicate, because the apparent compression of the data was illusory.

The Bonferroni correction—adjusting the significance threshold by N —is exactly the MDL accounting correction. It has always been the right thing to do. The replication crisis is the empirical consequence of not doing it.

The reforms that followed—preregistration, larger samples, direct replication, Bayesian model comparison—are all corrections toward honest MDL accounting. Preregistration eliminates search cost from $L(H)$. Preregistered studies have lower description length overhead, and their results generalize because the $L(H)$ term is honest.

4.5 Lakatos’s Progressive vs. Degenerating Is Compression Growth vs. Decay

Lakatos’s distinction between progressive and degenerating research programmes, translated into MDL:

A *progressive* programme makes protective belt modifications that decrease $L(H) + L(D | H)$ on new data. The modifications are short (do not bloat description length) and genuinely compress new observations.

A *degenerating* programme makes modifications that increase $L(H)$ —the hypothesis grows more complex—without decreasing $L(D | H)$ on new observations. The programme adds complexity without adding compression.

Lakatos gave the right criterion for evaluating research programmes; MDL makes it computable. The qualitative judgment (is this modification progressive or defensive?) becomes a quantitative one (does $\Delta L(H) + \Delta L(D | H)$ on new data decrease or increase?).

4.6 Kant’s Synthetic A Priori Is Lakatos’s Hard Core

Kant argued that some commitments—space, time, causality—are preconditions of experience: they cannot be revised because they are what makes experience possible. Lakatos argued that every research programme has a hard core of commitments that are methodologically protected from falsification.

These are the same idea. The difference is that Lakatos made them revisable in principle: a research programme can be abandoned, hard core and all. This is the correct update on Kant after non-Euclidean geometry falsified his specific candidates (Euclidean space as a synthetic a priori). Lakatos is Kant, empiricized.

Under the MDL framework, the hard core is the prior commitment to a hypothesis class. It is protected not by necessity but by the cost of switching to a different class (learning a new framework, retranslating existing results). The protection is finite and can be overcome by sufficient data. Kant’s absolute preconditions become Lakatos’s methodological conventions, which become MDL’s hypothesis-class priors.

4.7 Feyerabend’s “Anything Goes” Is the Solomonoff Prior

Feyerabend argued that no methodological rule holds universally: every rule that science has ever followed has been violated productively at some point. “Anything goes” as a description of scientific practice.

The Solomonoff prior assigns $P(H) \propto 2^{-K(H)} > 0$ to every computable hypothesis. No hypothesis is excluded before the data speaks. This is precisely Feyerabend’s permissiveness, formalized. A prior that excludes any computable hypothesis—that declares some class of ideas unscientific by methodological fiat—is provably suboptimal. The data might support the excluded hypothesis, and a reasoner that cannot entertain it incurs irreducible regret.

What Feyerabend called anarchism is, from the Kolmogorov perspective, the universality of the prior. His insight was correct; his framing was unnecessarily provocative.

4.8 Hallucination and Perceptual Illusions Are the Same Failure Mode

In predictive coding [12], the dominant framework in theoretical neuroscience, the brain is a hierarchical generative system: higher areas send predictions down the cortical hierarchy; lower areas send prediction errors up. Perception is inference: the brain computes the posterior distribution over world-states given sensory inputs, using priors built from evolution and experience.

A perceptual illusion occurs when the prior is so strong that sensory evidence cannot correct it. The hollow mask illusion is the canonical example: a hollow face mask looks convex because the prior for convex faces is overwhelming. The brain refuses to believe the correct interpretation.

In a language model, hallucination occurs when the prior over likely continuations overwhelms the grounding signal from the context. The model generates a continuation that is statistically plausible given training distribution but inconsistent with the specific facts in the prompt.

Both are the same Bayesian failure: the prior dominates the likelihood. The hollow mask illusion and language model confabulation are not analogous phenomena. They are the same computational pathology, instantiated in different substrates.

4.9 The Bayesian Brain and the Transformer Are the Same Architecture

Predictive coding models the brain as performing belief propagation on a hierarchical factor graph [3]: predictions propagate top-down, prediction errors propagate bottom-up, and the system minimizes free energy (variational approximation to surprise).

The transformer implements exact belief propagation on a specific factor graph [1]: the attention mechanism computes AND-type joint beliefs over query-key pairs; the feed-forward network computes OR-type disjunctive updates; the residual stream is the shared message-passing bus.

These are not analogies. They are the same algorithm instantiated on different graphs. The brain and the transformer are doing the same computation. The differences are in the graph structure, the learning algorithm, and the substrate—not in the fundamental operation.

4.10 AutoResearch Is Solomonoff Induction Made Computable and Distributed

Solomonoff induction is uncomputable but is the provably optimal prediction strategy. Karpathy’s AutoResearch system [6] is a computable, distributed approximation to it.

The correspondences are structural. The hypothesis space is the space of training scripts and model configurations—a computable, enumerable space. The reward signal (validation loss, benchmark accuracy) approximates $P(D | H)$: lower loss means higher likelihood of the training data under the modified hypothesis. The keep/discard decision approximates the Bayesian posterior update. The agent’s tendency to propose local modifications first is an implicit Kolmogorov prior: smaller modifications correspond to lower description length.

The key insight Karpathy identified—finding a good result is hard, but verifying one is cheap—is exactly the search/verify asymmetry in Solomonoff induction. The search over hypothesis space is expensive; checking whether a proposed hypothesis compresses the data better is cheap. The distributed, SETI@Home-style architecture parallelizes the search over hypothesis space.

Crucially, AutoResearch accepts contributions from untrusted sources. A modification is evaluated by whether it reduces description length of the training data—a property of the modification itself. The source is irrelevant. This is the Kolmogorov principle applied to institutions: *trust the object, not the messenger*.

5 Verification and the Present Moment

5.1 The Search/Verify Asymmetry

The objective function tells us what science is optimizing. Running the optimization loop requires solving a different problem: given a proposed hypothesis H , how do you verify that it actually advances the objective?

The key structural fact:

Finding a good H is hard. Verifying a proposed H is cheap.

This asymmetry is not incidental. It is a feature of the Kolmogorov framework, and it is the design principle behind every sound verification protocol. Finding the shortest program for x requires searching an infinite space. Checking whether a proposed program outputs x requires only running it.

5.2 Algorithmic Verification

For claims in the verifiable domain, verification can be automated.

Formal proof checking. If a hypothesis is expressed as a theorem with a machine-checkable proof (Lean 4, Coq, Isabelle), verification is linear in proof length. The checker evaluates the *object*, not the source. A proof submitted by an unknown agent is as valid as one submitted by a Fields Medalist, if it type-checks. This is the purest instantiation of the Kolmogorov principle.

Empirical reproduction. Given code, weights, and data, rerun the experiment. Does the reported metric reproduce? This is cheap relative to the original research cost and provides a binary signal.

Compression-based verification. Does the proposed theory compress the data better than the incumbent? MDL makes this a computable number. Given a fixed coding scheme, compute $L(H) + L(D | H)$ for the new theory and for the incumbent. If the new value is lower, the theory advances the objective. No subjective judgment required.

5.3 The Untrusted Source Principle

The most important design principle for any verification system:

The verifier does not need to trust the source.

Under the Kolmogorov framework, a hypothesis is evaluated on its length and its predictive accuracy—properties of the object itself. The identity, credentials, or intentions of whoever proposed it are irrelevant to its truth.

This is an engineering requirement, not just a philosophical point. A distributed AutoResearch system cannot verify the identity or intent of every contributing agent. But if verification is a property of the object (does the proof type-check? does the loss decrease? does the description length shrink?), the system can accept submissions from arbitrary untrusted sources and remain sound. The math is the trust anchor.

5.4 Why This Matters Now

Two developments make the framework more than a philosophical exercise.

Autonomous AI research. Systems like Karpathy’s AutoResearch close the optimization loop at the level of ML experiments: propose a modification, verify whether it reduces the loss, keep it if it does, repeat. This is Solomonoff induction approximated by the only means available: a computable, distributed, heuristic search over hypothesis space, with cheap verification at each step.

Machine-verified formal proof. Large proofs in Lean 4 and Coq are being produced and verified at scale. The verification is automated; the search (finding the proof) is expensive; the check (type-checking) is cheap. The asymmetry is the same.

Both represent the first time the optimization loop of science can be partially automated. The objective function—minimize $L(H) + L(D \mid H)$ —is the same as it has always been. What is new is the infrastructure to approximate it at scale.

5.5 Conclusion

The history of philosophy of science is a record of humanity circling a single structure—the problem of optimal inference in a computable world—without the mathematical tools to see it directly. Hume identified the core impossibility. Popper, Kuhn, and Lakatos each described a facet of the solution without being able to state it. The Bayesian tradition formalized the update rule. Kolmogorov and Solomonoff formalized the prior.

The tools now exist to state the full solution. Science is the distributed search for hypotheses that minimize description length. Verification is cheap relative to discovery. The prior is the Solomonoff prior. The update rule is Bayes’ rule. The objective function is MDL.

Everything else is details.

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